

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	27 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization

Step 1: Problem Statement

Selected Problem

Small and medium farmers face difficulties in making timely agricultural decisions because they lack real-time information about crop diseases, weather changes, irrigation scheduling, fertilizer recommendations, and market prices. Existing solutions are often expensive, difficult to understand, or unavailable in regional languages.

Step 2: Brainstorming

Idea Category 1 – Crop Monitoring

- AI detects crop diseases from leaf images
 - Pest identification using smartphone camera
 - Crop growth monitoring
 - Early disease alerts
 - Nutrient deficiency detection
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Idea Category 2 – Weather & Irrigation

- AI irrigation scheduler
 - Rainfall prediction
 - Soil moisture monitoring
 - Smart water management
 - Water-saving recommendations
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Idea Category 3 – Farmer Assistance

- Voice chatbot in regional languages
- Text chatbot

- Image-based crop diagnosis
- Personalized farming calendar
- Government scheme recommendations

Idea Category 4 – Market Intelligence

- Crop price prediction
- Best market recommendation
- Demand forecasting
- Harvest timing suggestions
- Nearby buyers information

Idea Category 5 – Farm Management

- Fertilizer recommendation
- Seed recommendation
- Crop rotation planning
- Yield prediction
- Expense tracking

Step 3: Idea Prioritization

Idea	Importance	Feasibility	Priority
Smart Crop Recommendation	Very High	High	☆☆☆☆☆
Soil Analysis	Very High	High	☆☆☆☆☆
Crop Suitability Prediction	High	High	☆☆☆☆
Environmental Analysis	High	High	☆☆☆☆
K-Means Clustering	High	Medium	☆☆☆☆
Logistic Regression	Medium	Medium	☆☆☆
Agricultural Dashboard	Medium	High	☆☆☆
Crop Prediction Web Application	Medium	Medium	☆☆☆

Final Selected Ideas

- Smart Crop Recommendation System
- Soil and Environmental Analysis
- Machine Learning-Based Crop Prediction
- Agricultural Decision Support System

These become the core modules of the project.